



Complex Social &
Computational Systems

WHAT IS MACHINE LEARNING AFTER ALL?

A PSYCHOLOGIST'S GUIDE TO MACHINE LEARNING
ZURICH, MARCH 16-20, 2026

3 MINUTES

BEFORE WE START:

**THINK ABOUT OR DISCUSS WITH YOUR NEIGHBORS
MACHINE LEARNING VOCABULARY THAT YOU KEEP
ON HEARING BUT NEVER COULD MAKE SENSE OF IT.
WRITE IT DOWN — WE'LL GET BACK TO IT LATER.**

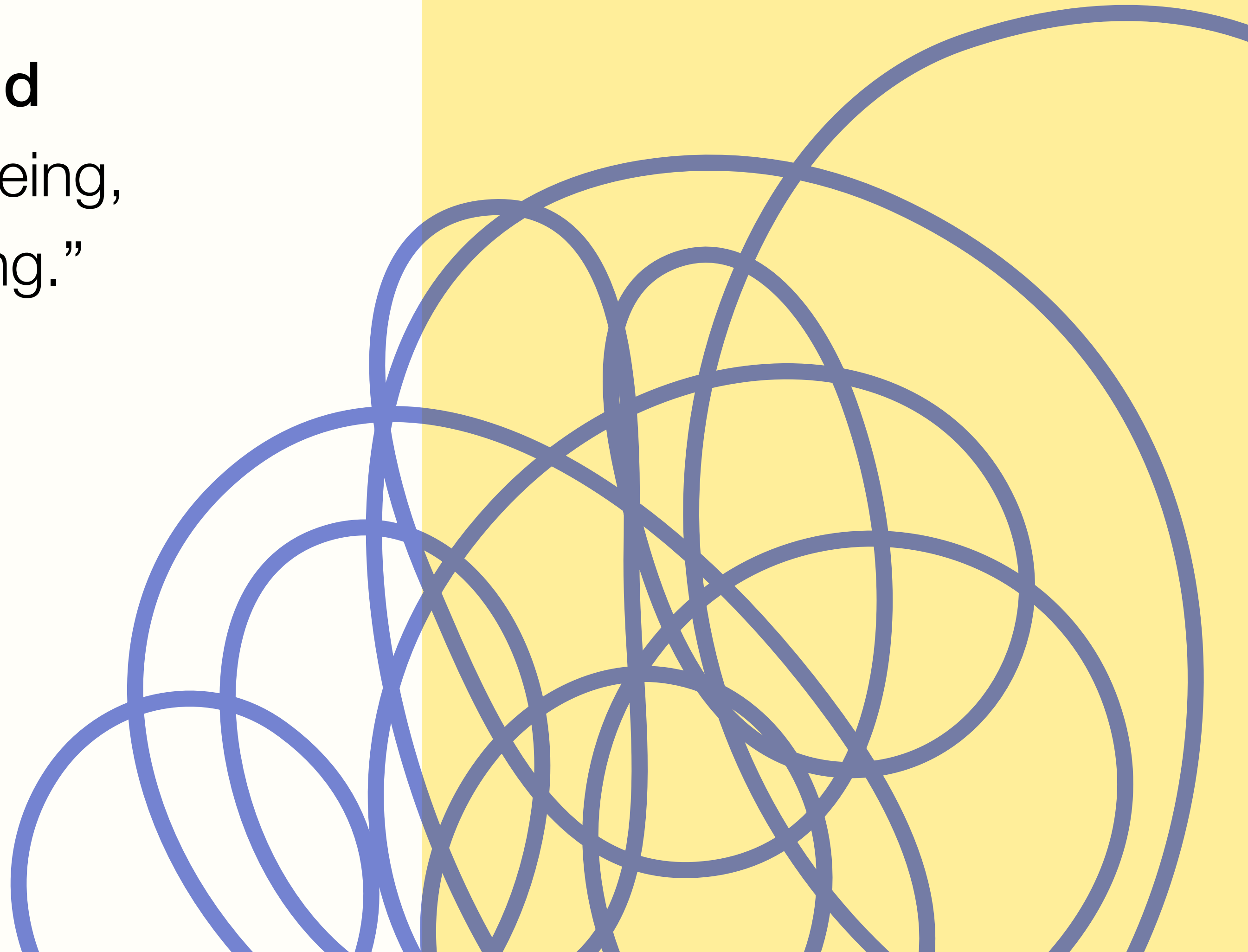
**NOWADAYS, EVERYONE IS
TALKING ABOUT
“ARTIFICIAL INTELLIGENCE”.**



AI: MORE AN OPINION THAN A DEFINITION.

"... technologies that **complement and strengthen human capabilities** in seeing, hearing, analyzing, deciding, and acting."

(Microsoft, What is artificial intelligence?
Definitions & functions of AI)



AI: MORE AN OPINION THAN A DEFINITION.

"... a **field of research in computer science** that develops and studies methods and software that **enable machines** to perceive their environment and **use learning and intelligence** to take actions that maximize their chances of **achieving defined goals.**"

(Russel & Norvig, Artificial Intelligence:
A Modern Approach)



AI: MORE AN OPINION THAN A DEFINITION.

"... the ability of a **machine to display human-like capabilities** such as reasoning, learning, planning and creativity."

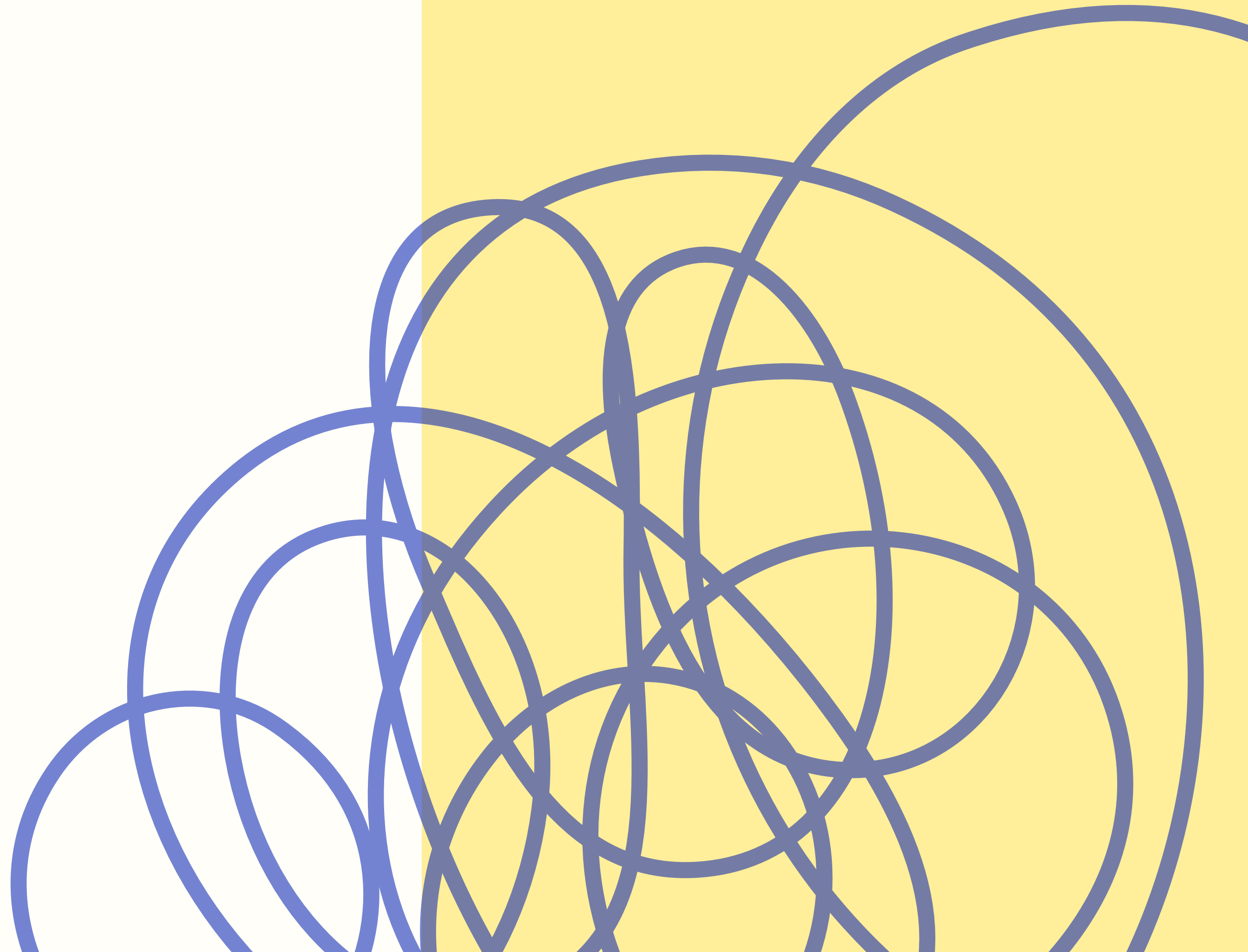
(European Parliament, What is artificial intelligence and how is it used?)

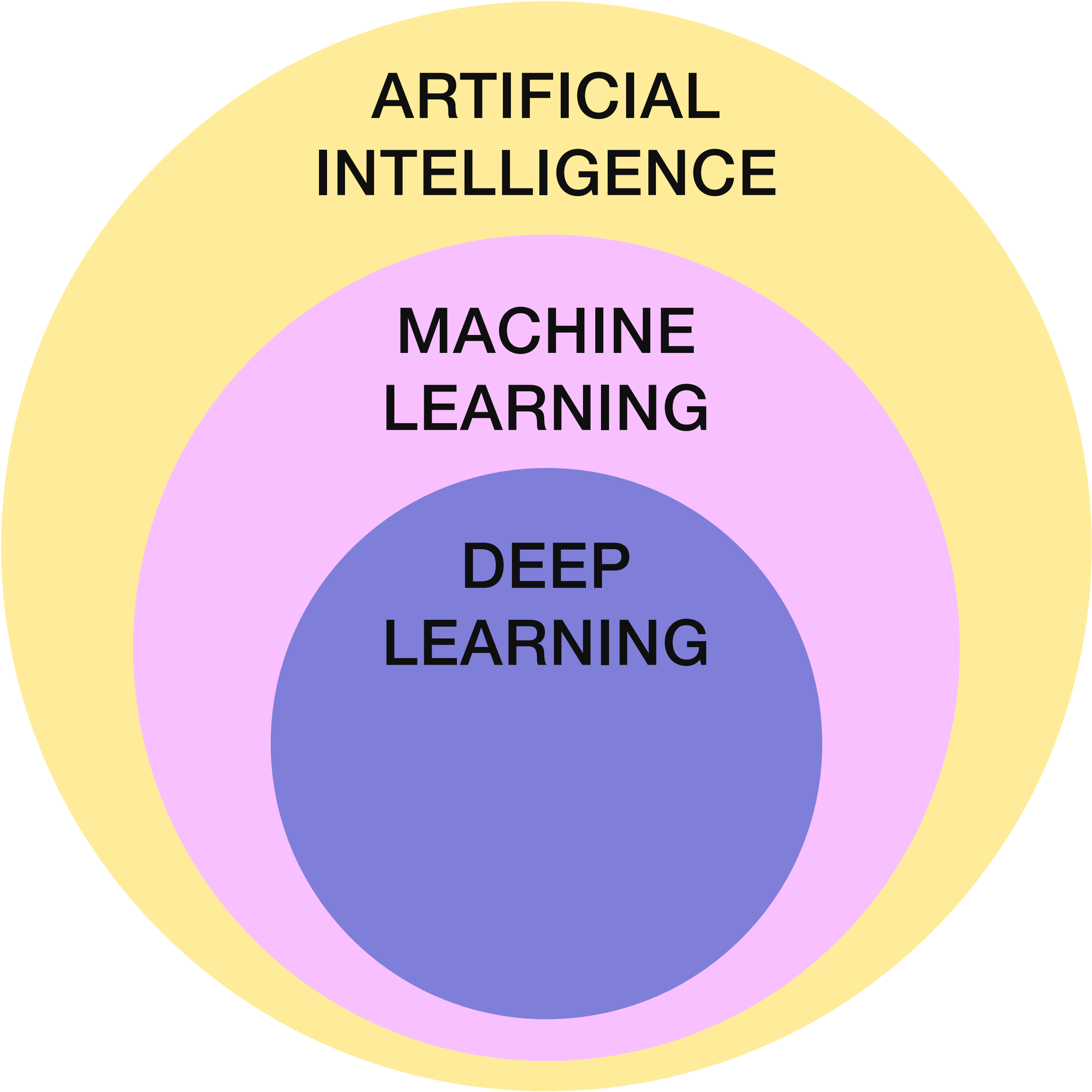


AI: MORE AN OPINION THAN A DEFINITION.

"... a **poor choice of words** in 1954."

(Someone on Twitter)



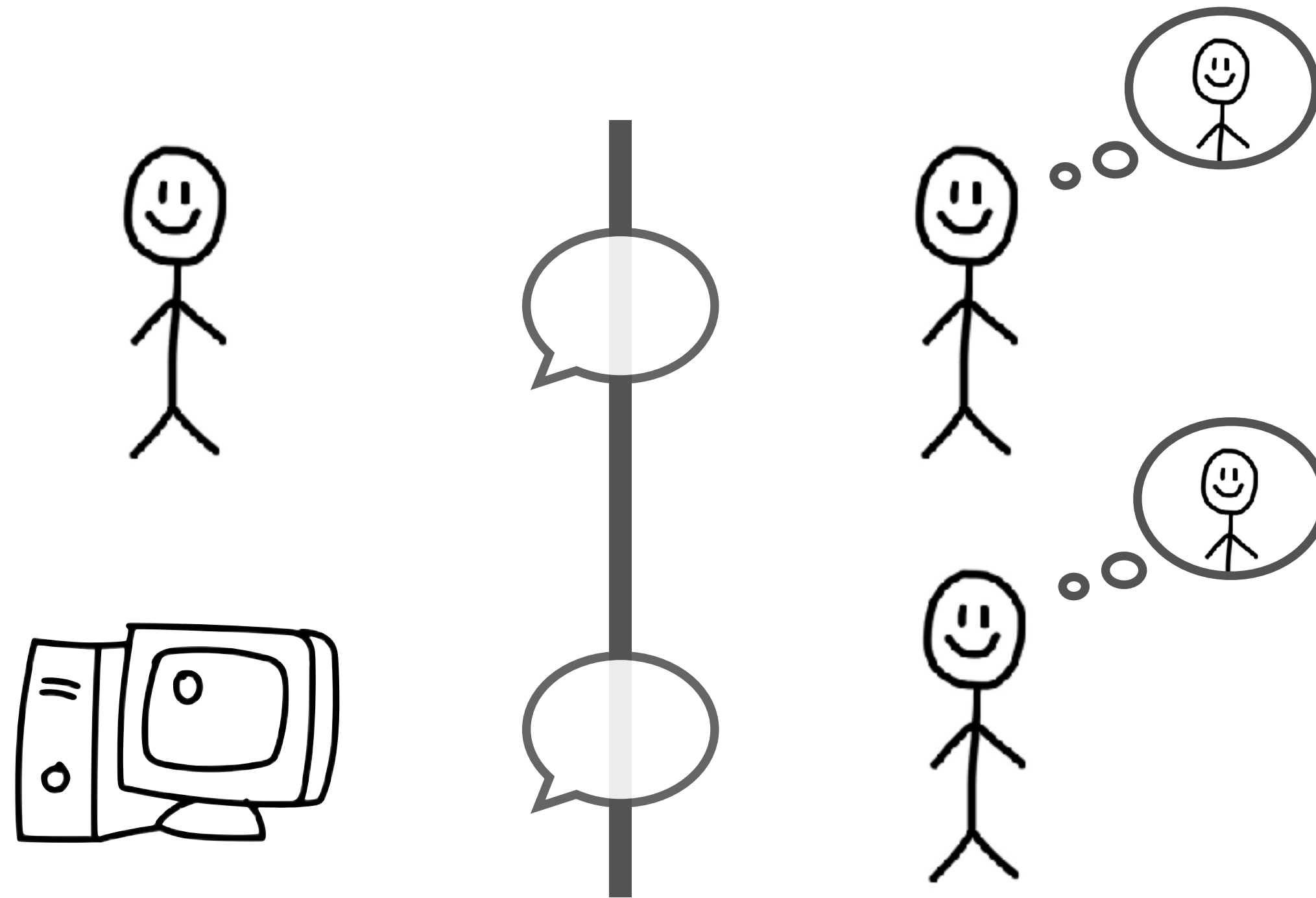


**ARTIFICIAL
INTELLIGENCE**

**MACHINE
LEARNING**

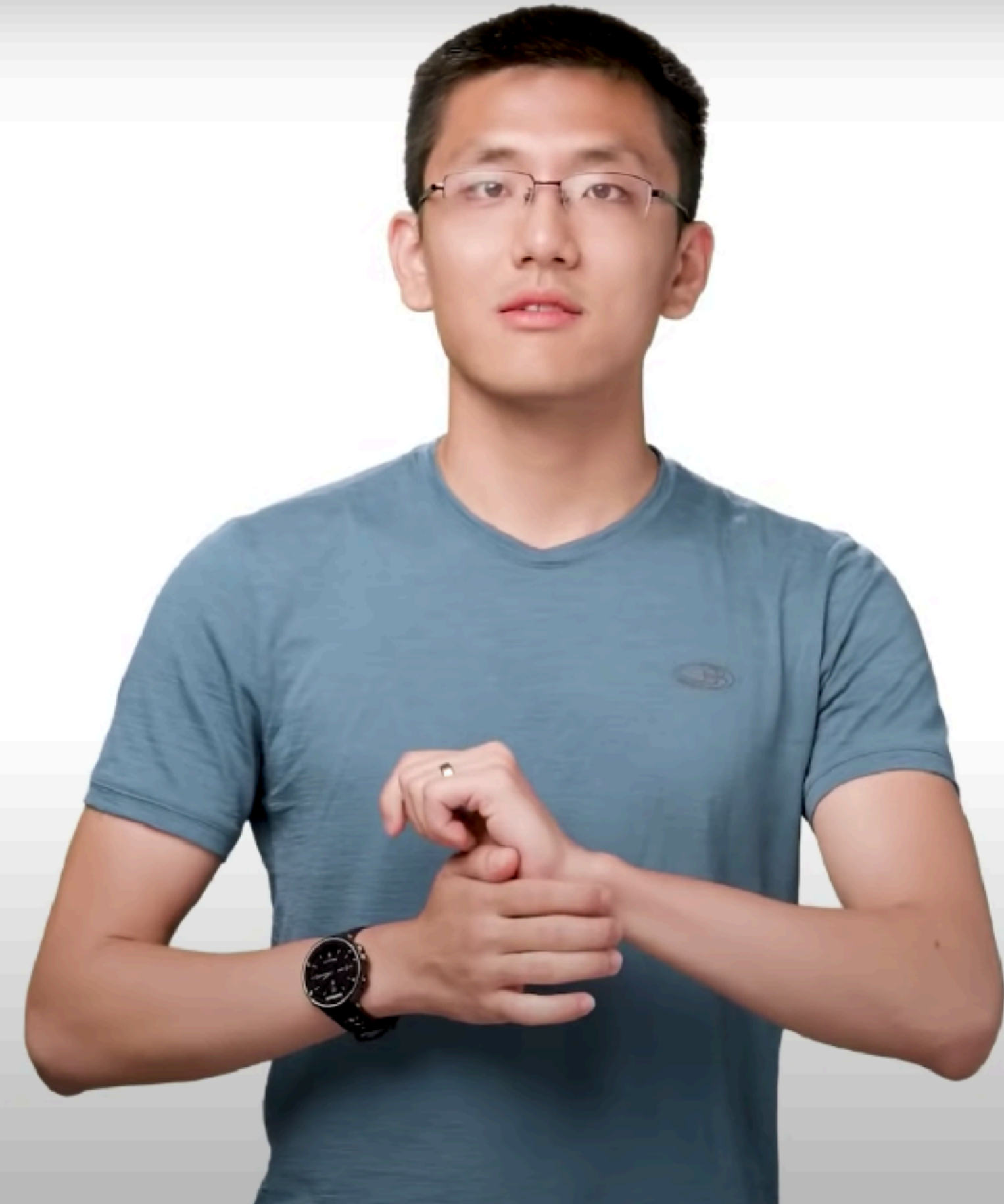
**DEEP
LEARNING**

ARTIFICIAL INTELLIGENCE – WHAT IS THIS? THE TURING TEST.



MACHINE LEARNING – WHAT IS THIS?

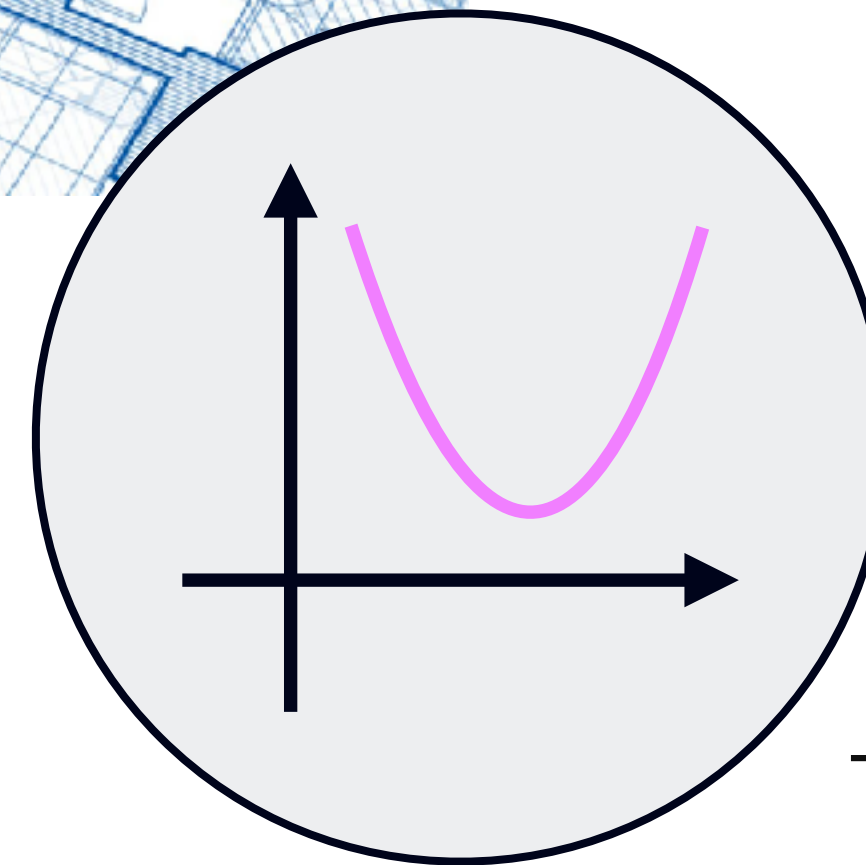
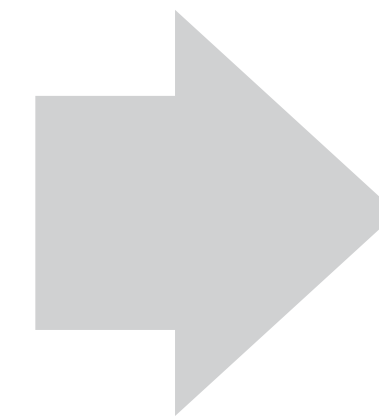
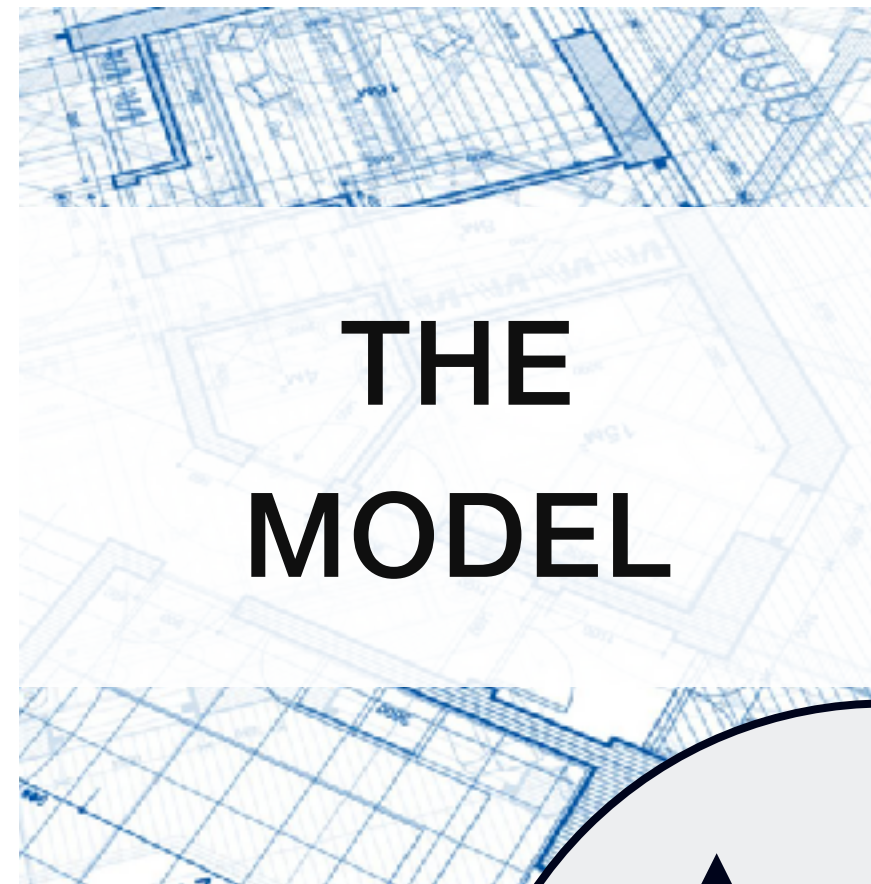
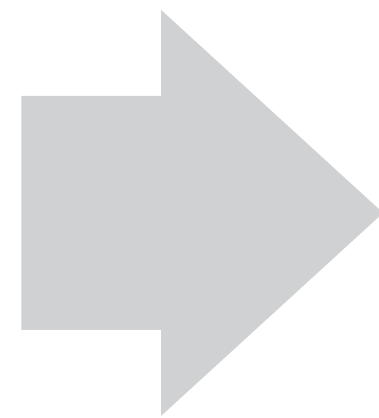
*Machine Learning is
using **data** to
answer questions*



MACHINE LEARNING – WHAT IS THIS?

0100100101101110011101
000110010101110010011
011100110000101110100
0110100101101110110111
00110000101100001
0000001001001010111
00110100011001010111
001001101110011001010
1110100001000000100
01000110000101111001

**THE
DATASET**

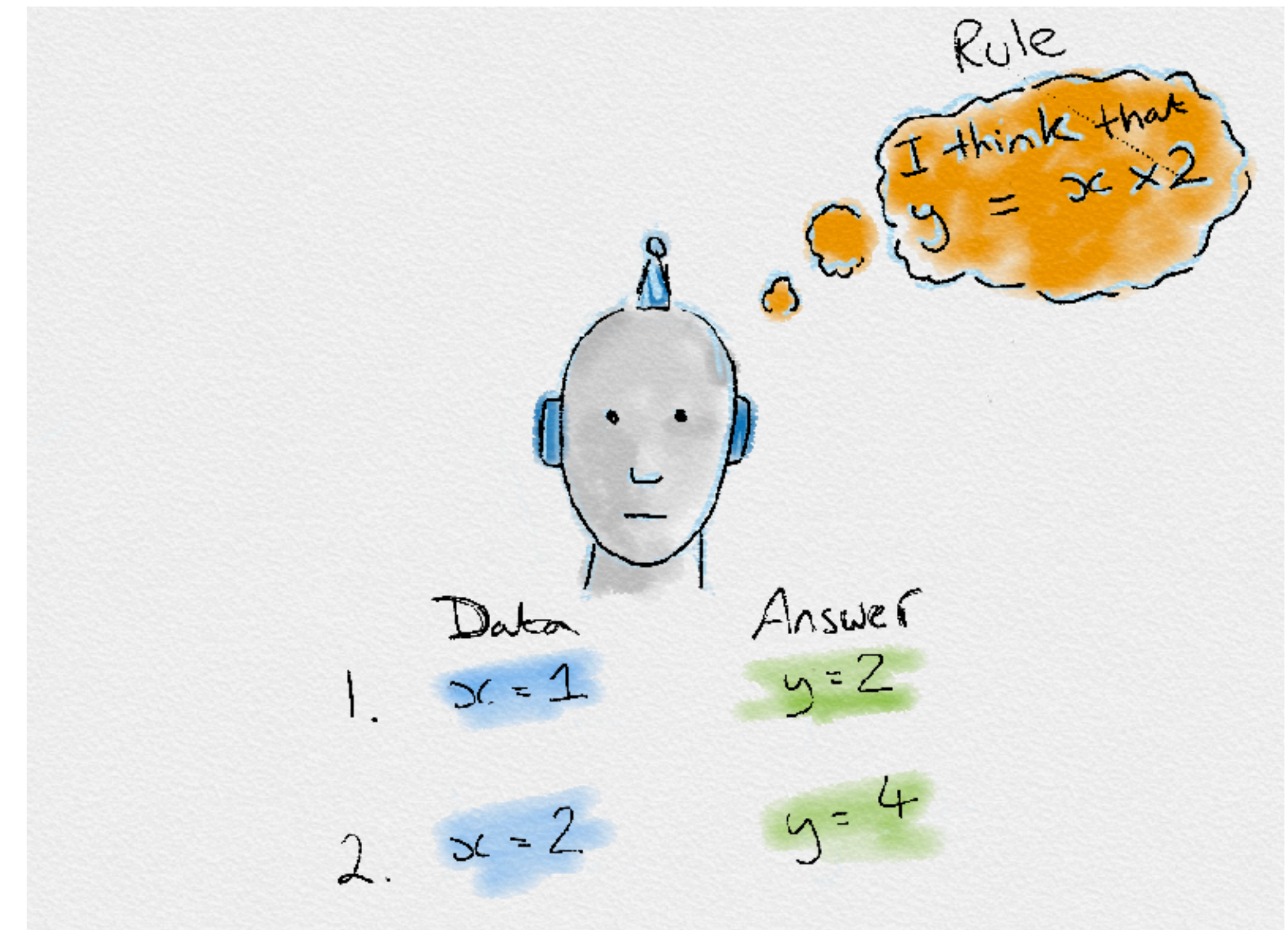
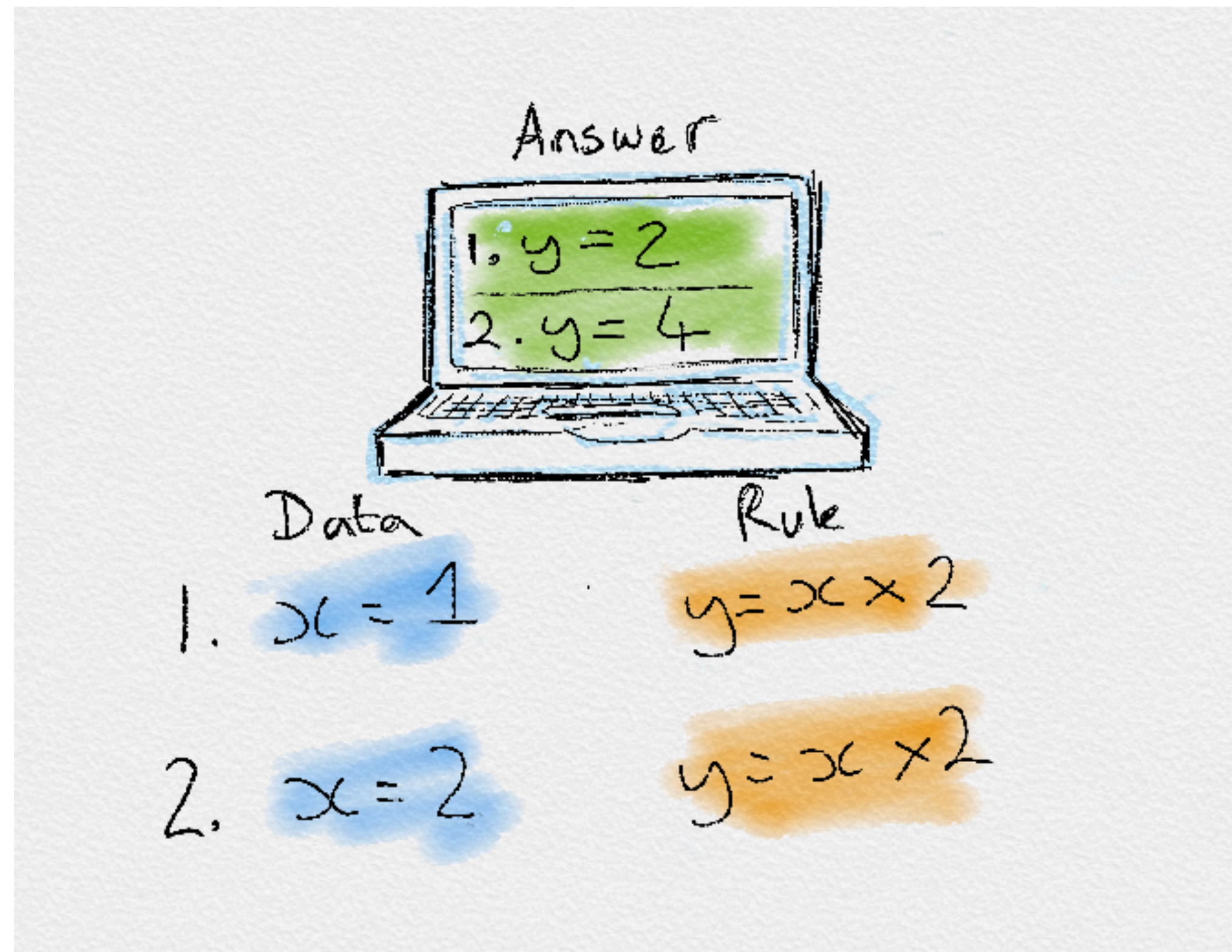


THE LOSS FUNCTION

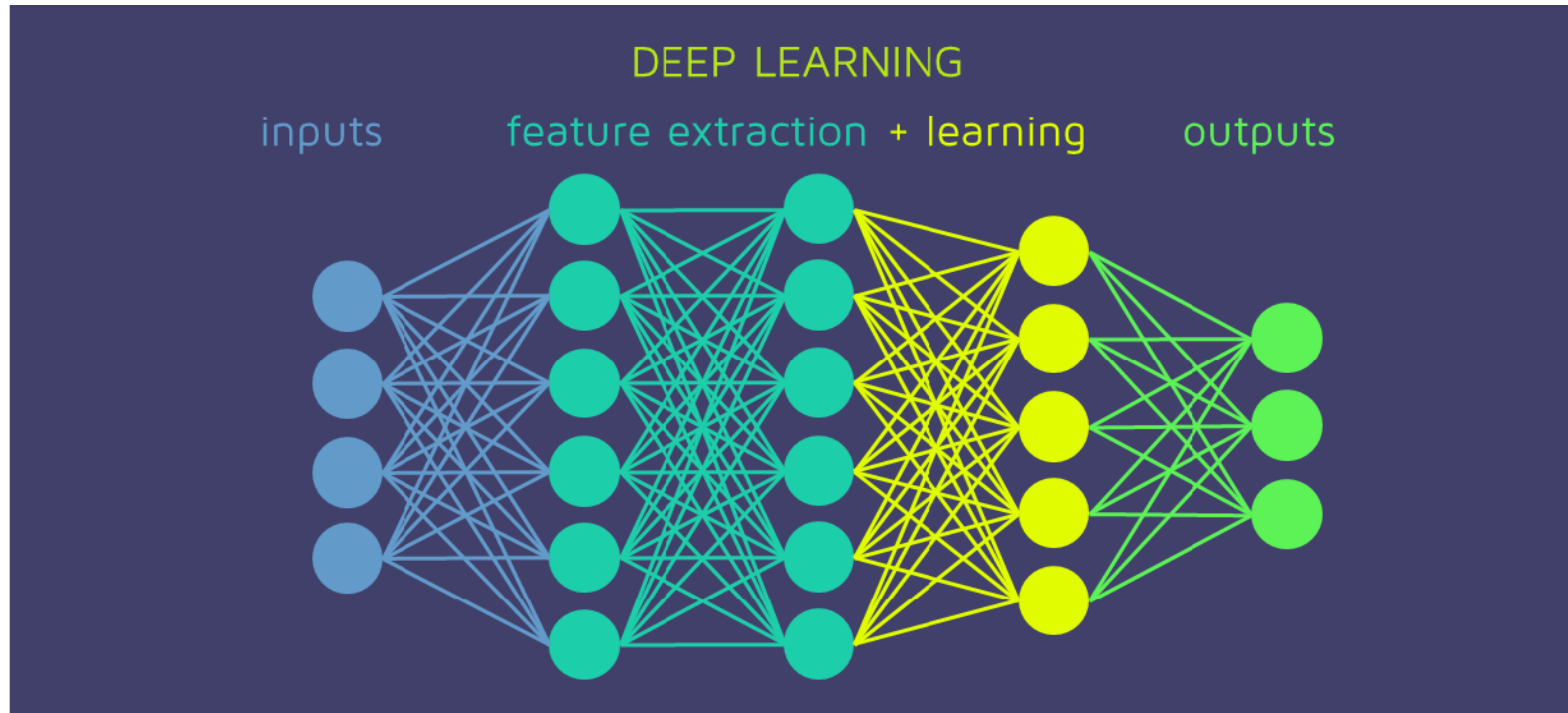
A photograph of a massive landfill filled with various types of waste, including plastic bags, cardboard, and other debris. In the background, a yellow bulldozer is visible, working on the pile. The sky is overcast with grey clouds. A semi-transparent white rectangular box is centered over the image, containing the text "GARBAGE IN. GARBAGE OUT." in bold black letters.

**GARBAGE IN.
GARBAGE OUT.**

MACHINE LEARNING – WHAT IS THIS?

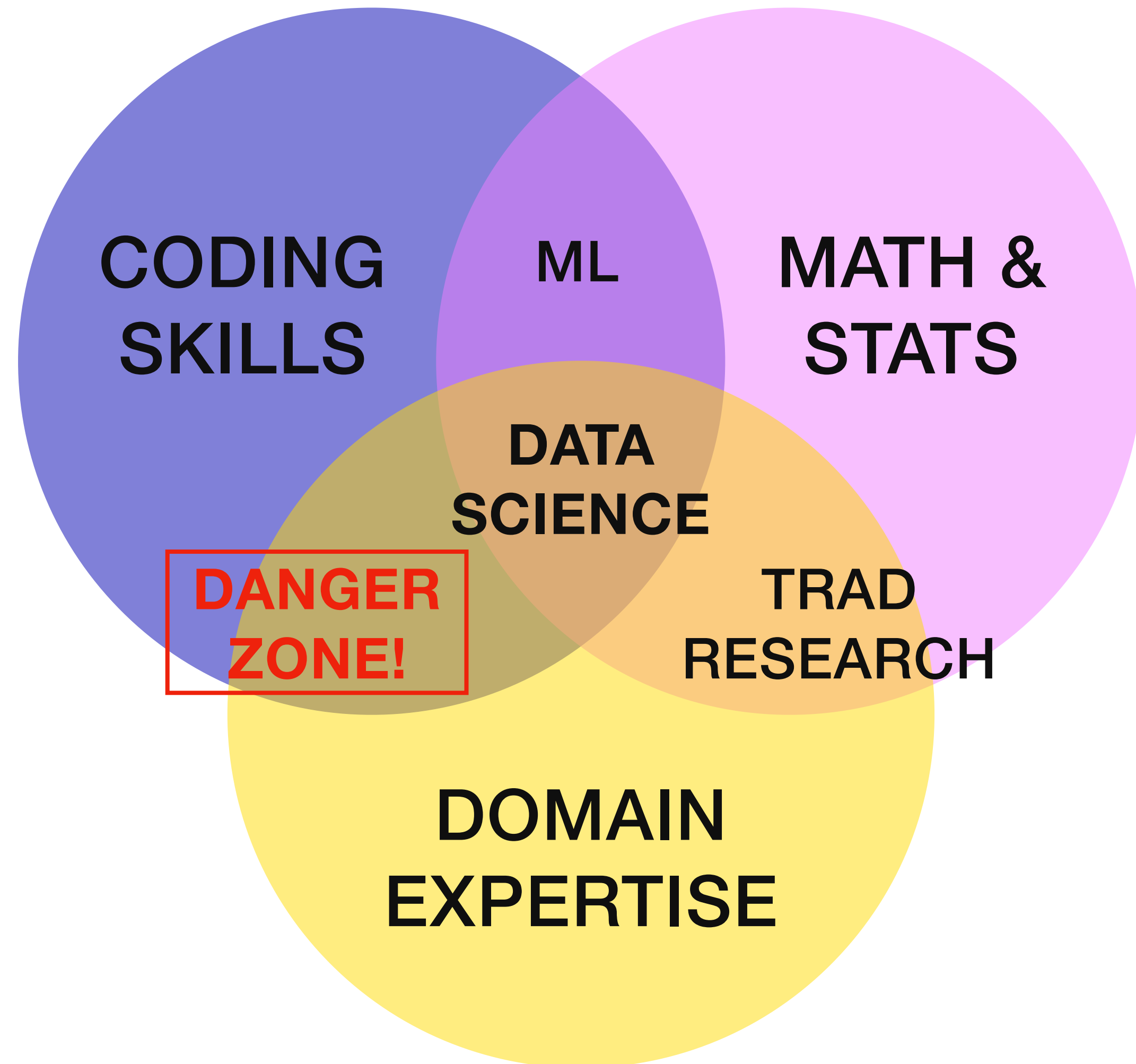


DEEP LEARNING – WHAT IS THIS?



EXCURSUS: DATA SCIENCE — WHAT IS THIS?

Data Science is the process of gaining knowledge from (unstructured) data.

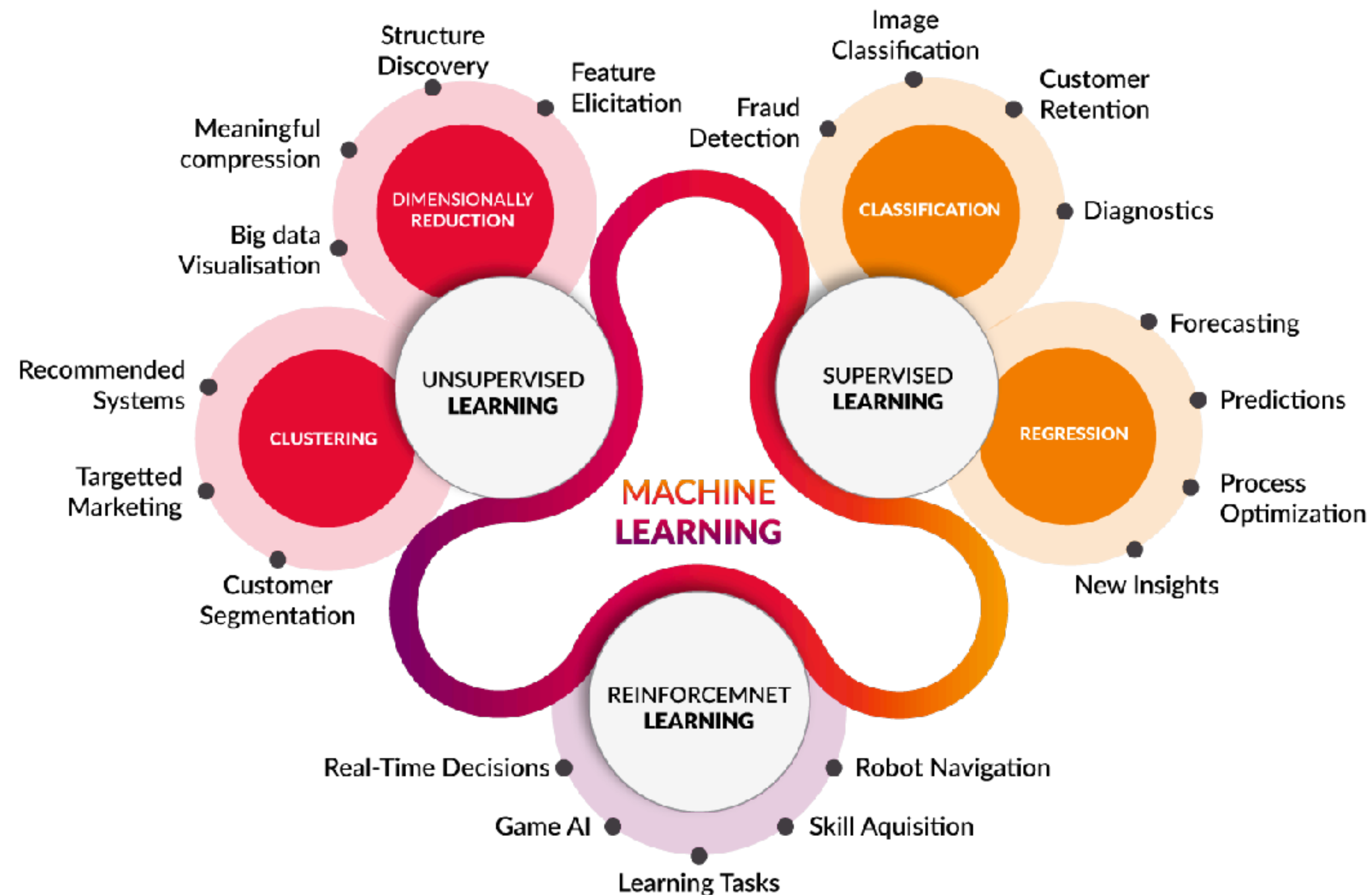


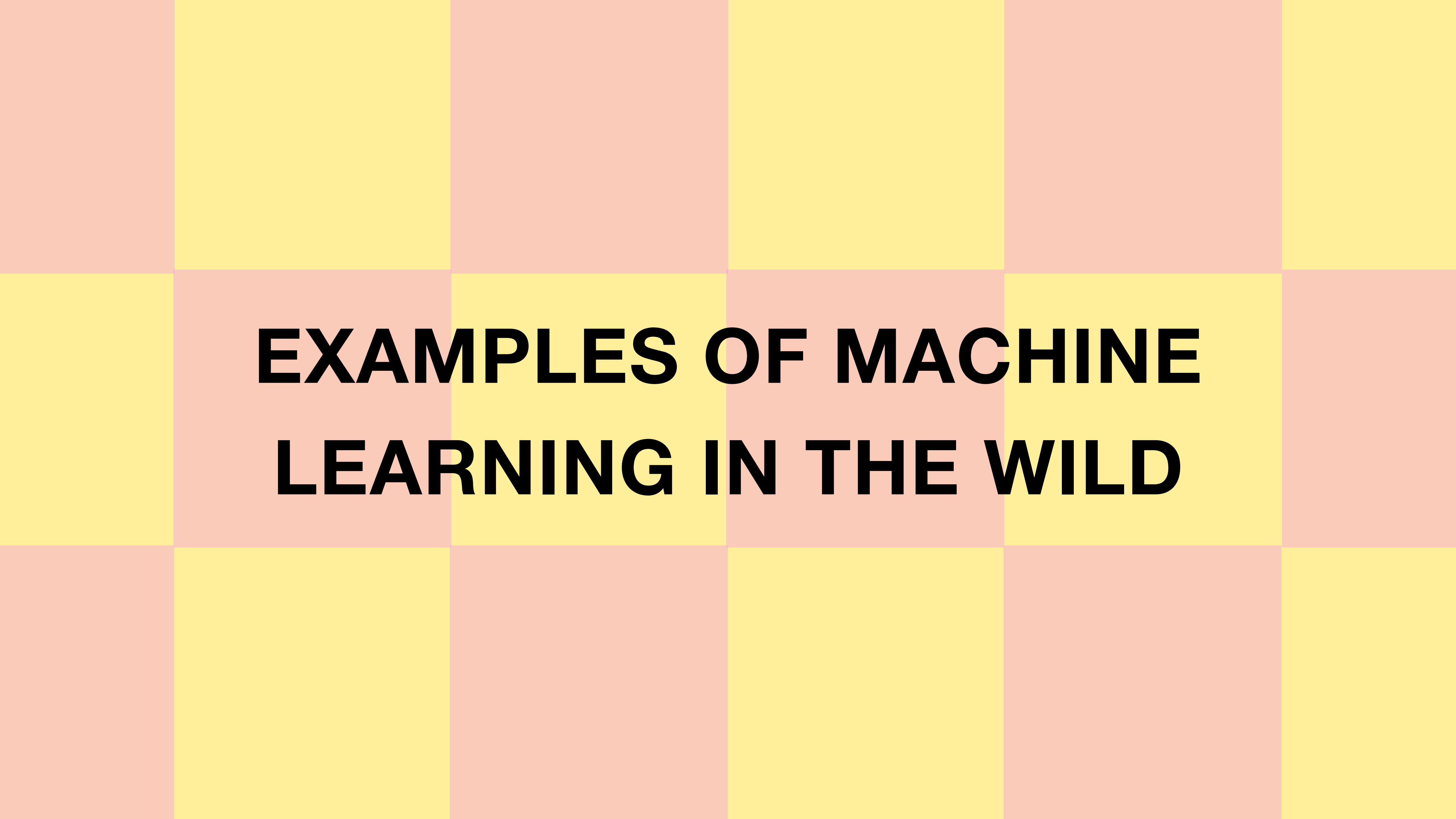
NO FREE LUNCH!

THEOREM



WHAT TYPES OF MACHINE LEARNING ARE THERE?





EXAMPLES OF MACHINE LEARNING IN THE WILD

NETFLIX ORIGINAL

STRANGER THINGS

95% Match 2017 2 Seasons 4K Ultra HD 5.1

When a young boy vanishes, a small town uncovers a mystery involving secret experiments, terrifying supernatural forces and one strange little girl.

Winona Ryder, David Harbour, Matthew Modine
TV Shows, TV Sci-Fi & Fantasy, Teen TV Shows

RECOMMENDER SYSTEMS

Popular on Netflix

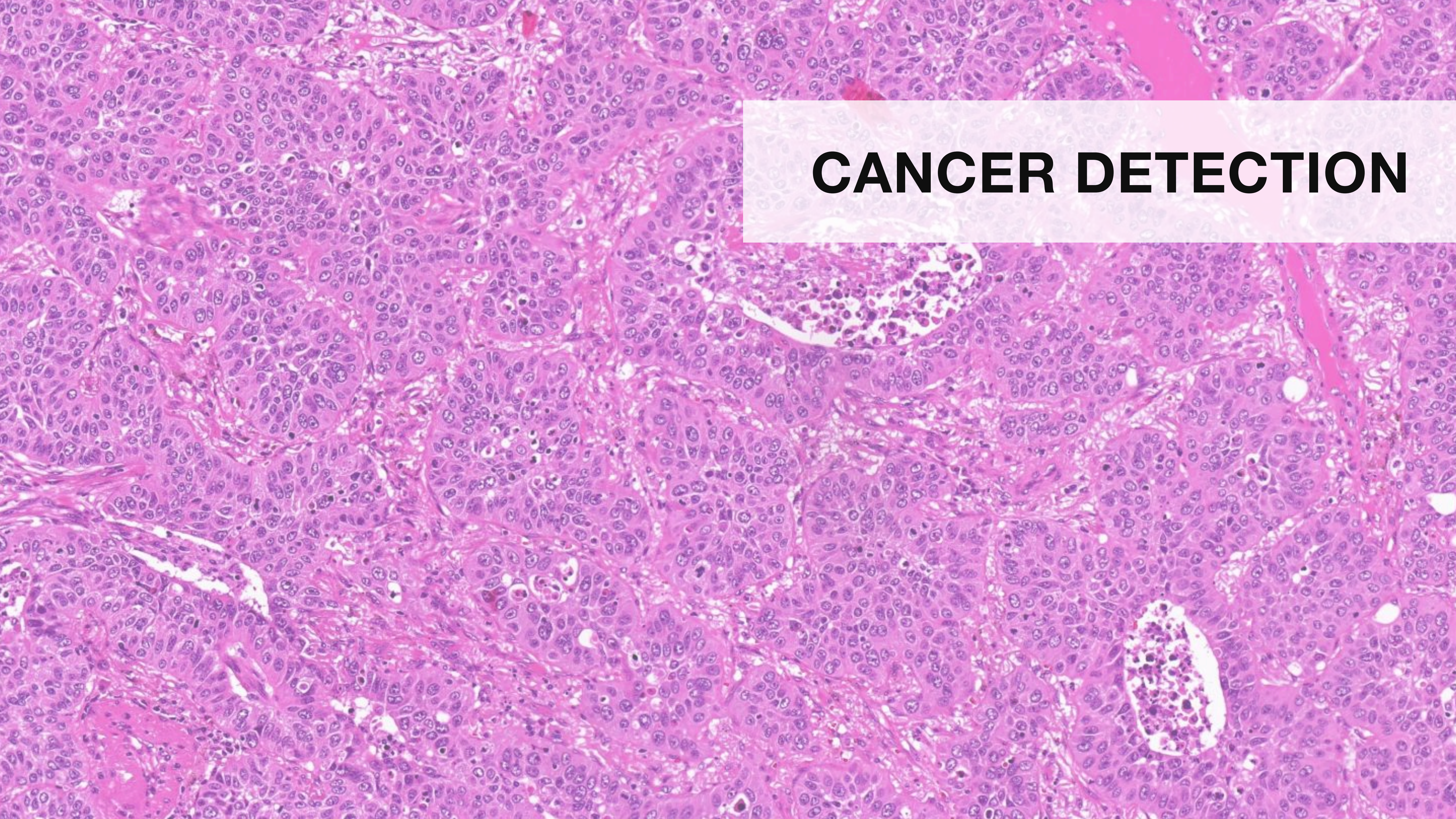


Recently Watched



FACE RECOGNITION



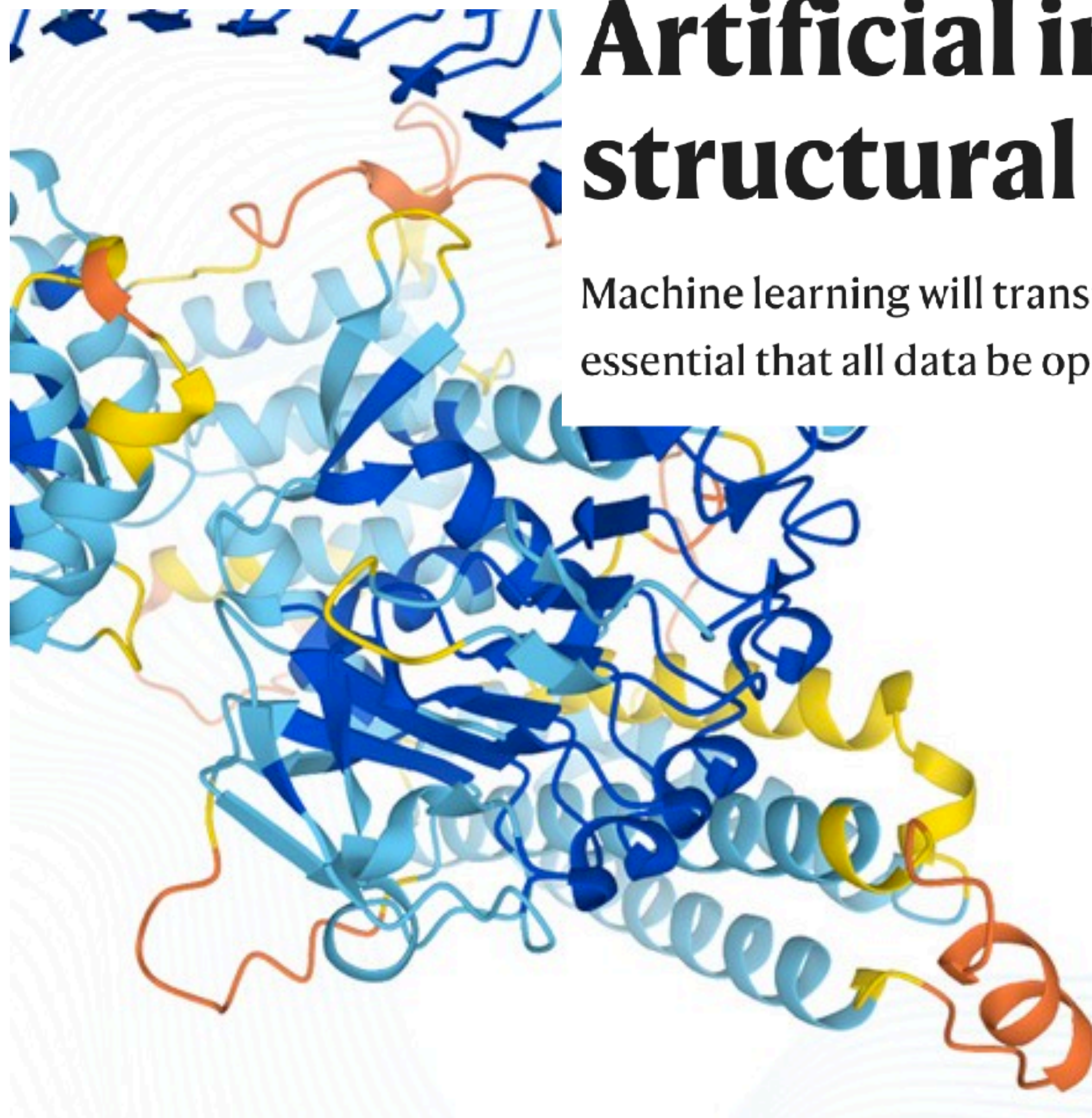
A high-magnification histological image of breast tissue stained with hematoxylin and eosin (H&E). The image shows a dense population of cells with dark purple nuclei and pink cytoplasm/extracellular matrix. There are several irregular, glandular-like structures (ducts) visible, some of which are filled with a mass of cells, indicating invasive ductal carcinoma. The overall architecture is disorganized, with cells breaking through the normal ductal boundaries.

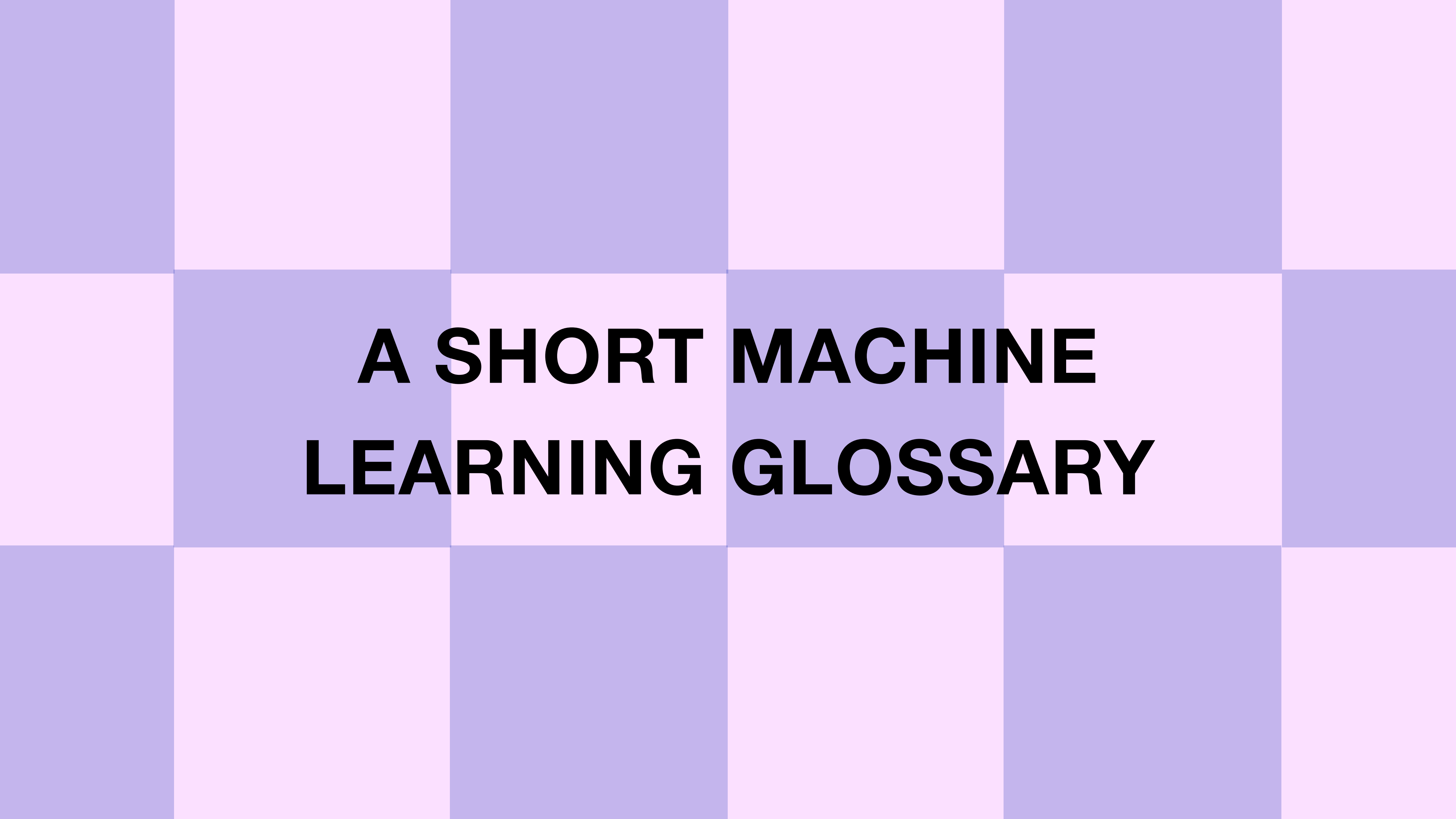
CANCER DETECTION

EDITORIAL | 27 July 2021

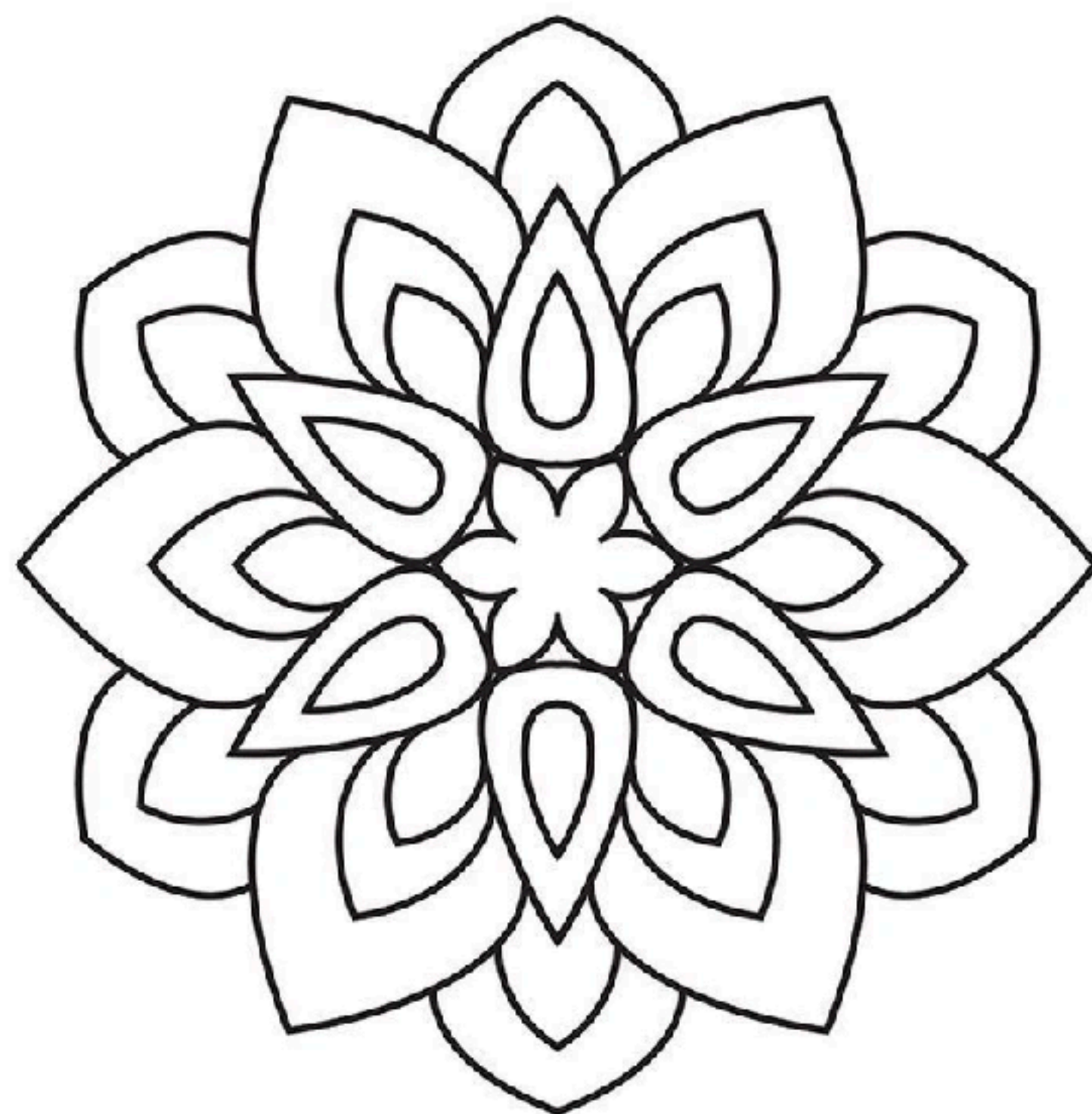
Artificial intelligence in structural biology is here to stay

Machine learning will transform our understanding of protein folding. And it's essential that all data be open.





A SHORT MACHINE LEARNING GLOSSARY



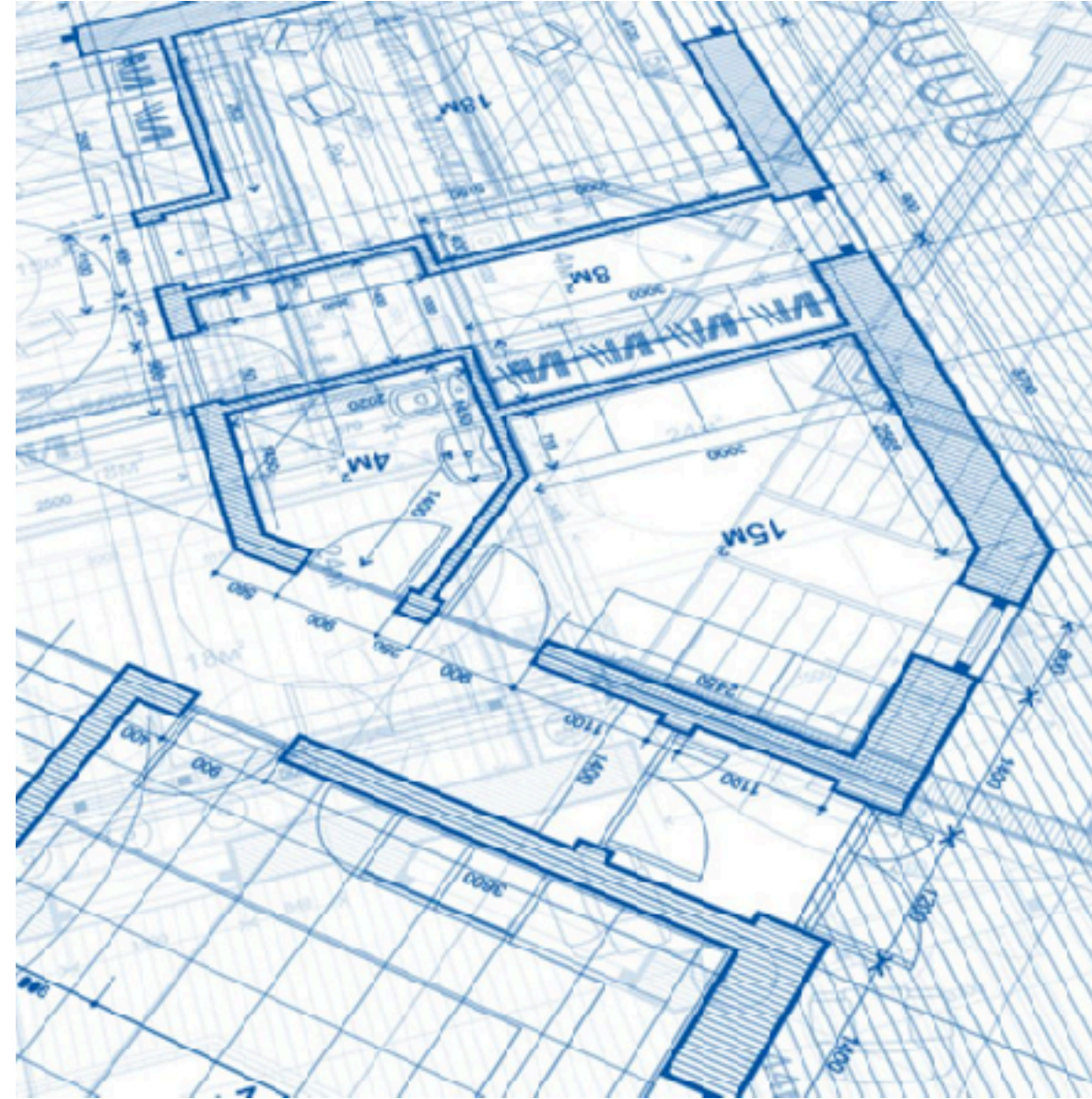
FEATURES & DIMENSIONS

$$2 + 2 = 4$$

GROUND TRUTH



GOLD STANDARD



MODEL



PARAMETERS



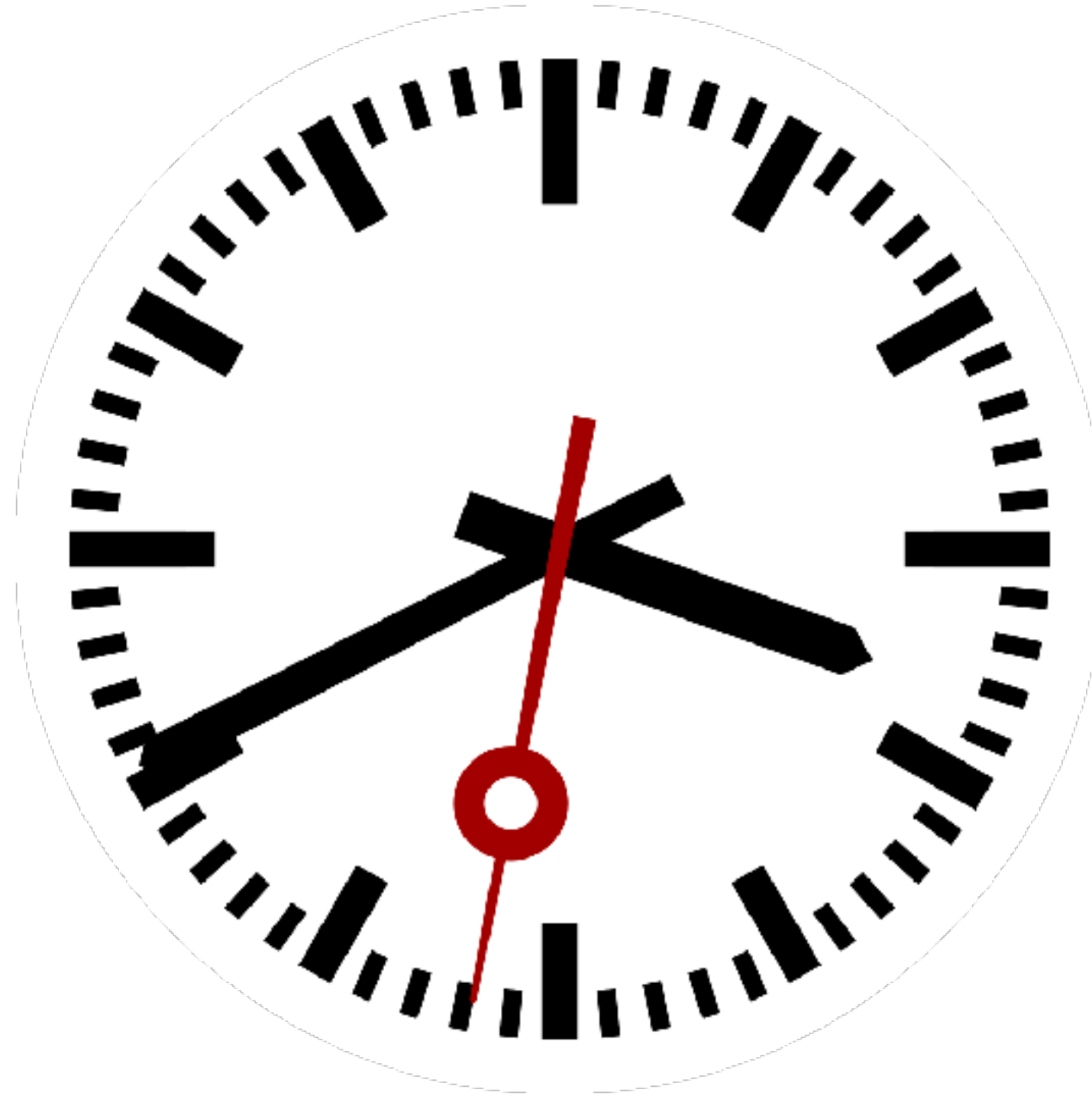
ARCHITECTURE



COST FUNCTION & LOSS



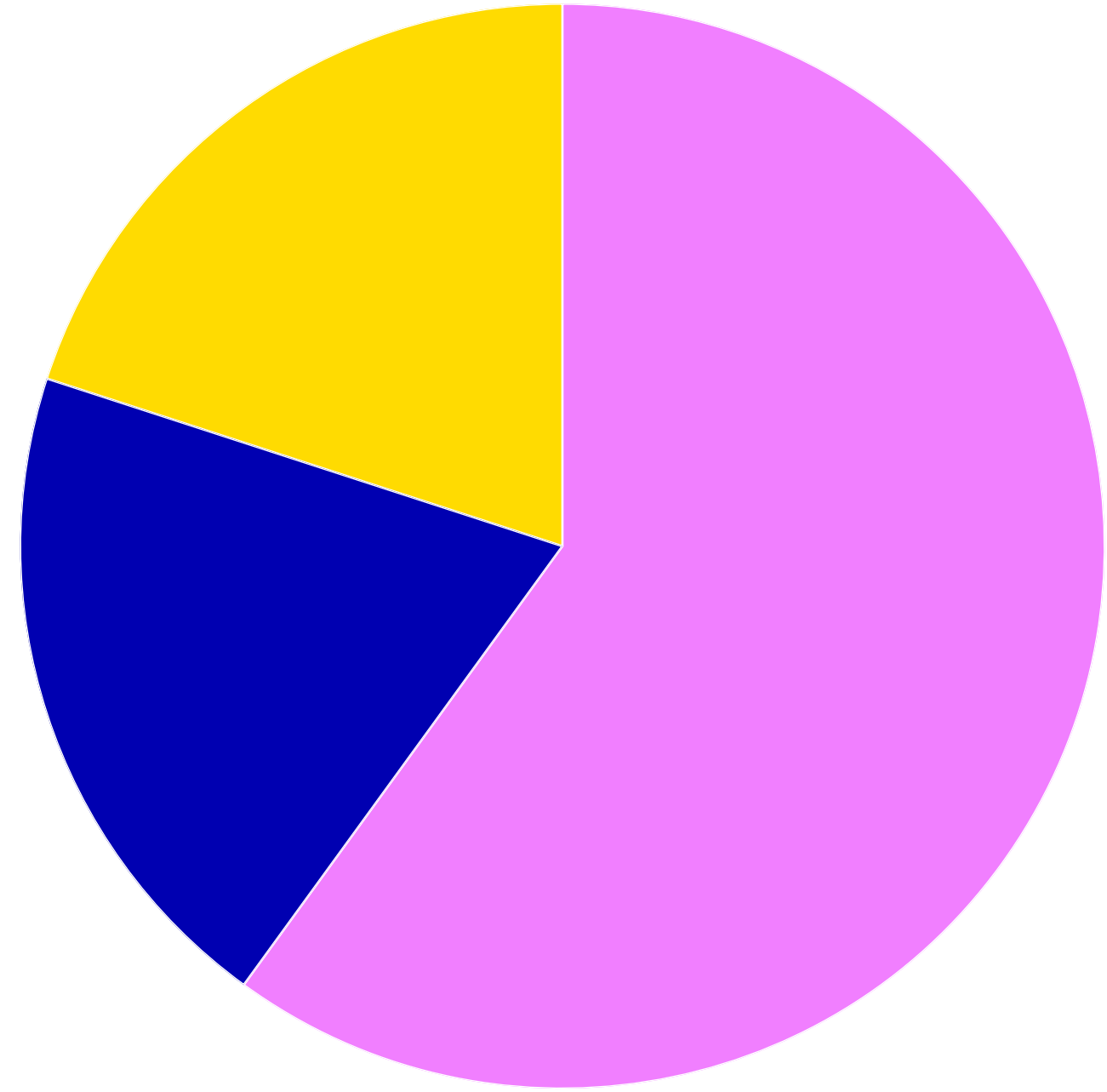
TASK



EPOCH



BATCH



TRAINING, VALIDATION & TEST SET



OVERFITTING & UNDERFITTING



TUNING



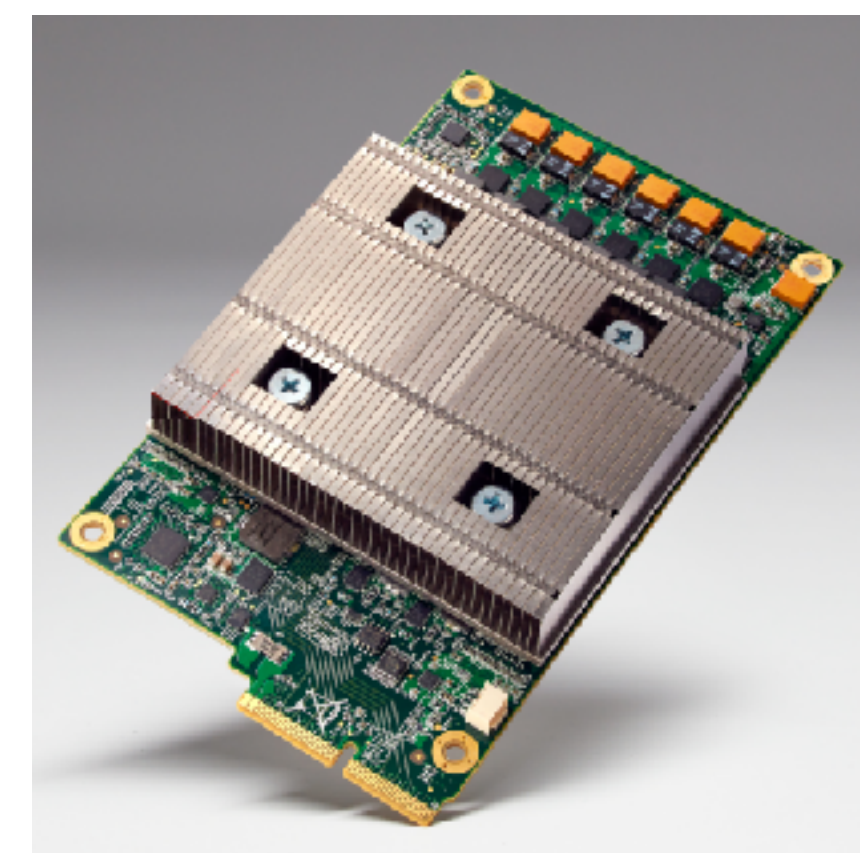
FINE-TUNING



PRE-TRAINING



TRANSFER LEARNING



CPU, GPU & TPU

**TELL ME WHAT YOU STILL
NEED TO KNOW.**

WHAT VOCABULARY DID I LEAVE OUT?

SOME VIDEO RECOMMENDATIONS

Computer Scientist Explains Machine Learning in 5 Levels of Difficulty

[a WIRED format]: <https://www.youtube.com/watch?v=5q87K1WaoFI&t=1370s>

Why It Was Almost Impossible To Solve Protein Folding (But AI Did) [a Veritasium video on AlphaFold]: https://www.youtube.com/watch?v=P_fHJIYENdI&t=2s

10:30 - 10:45
COFFEE BREAK